

## SERVICE COSTING

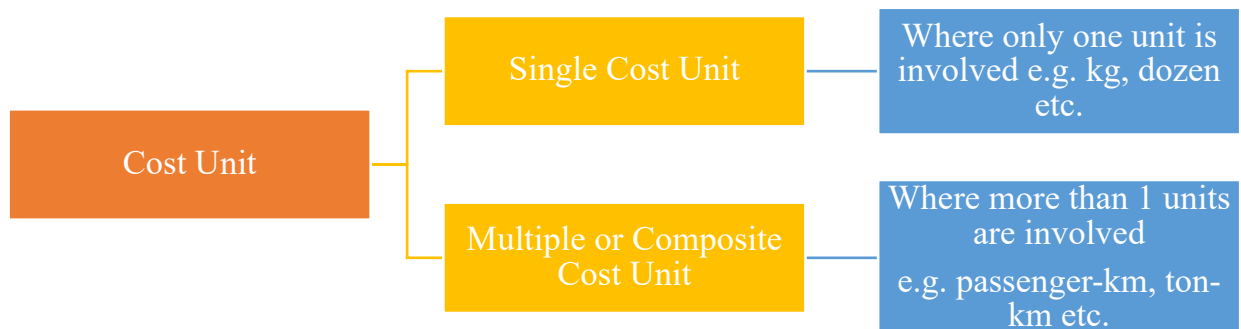
### 1. Service Costing

It can be internally or externally.

Industries will have high fixed cost than variable cost.

### 2. Cost Unit

The terms in which costs are expressed.



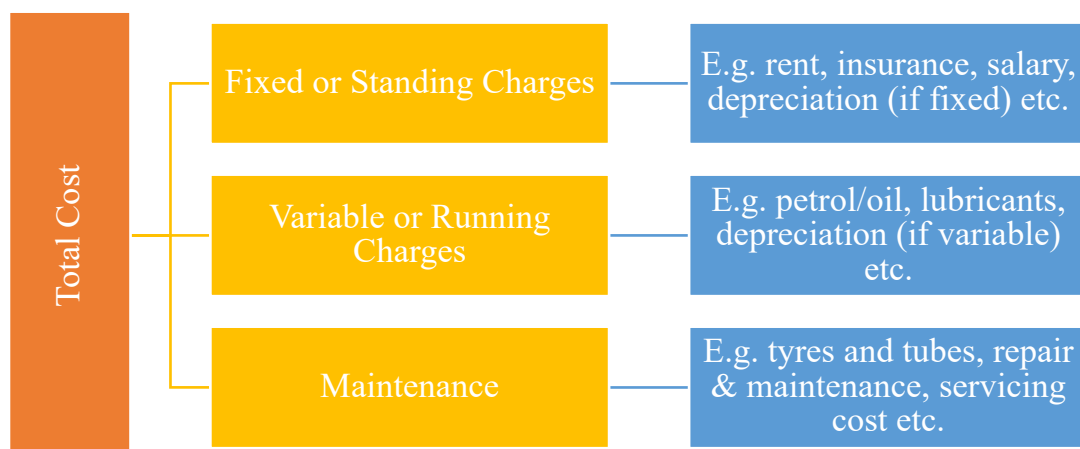
### 3. Cost Unit

(A) **Absolute Ton-km** = Actual kms × Actual tons

(B) **Commercial Ton-km** = Actual kms × Average tons

4. **Effective Cost Unit** = Total units – Normal loss units

### 5. Total Cost

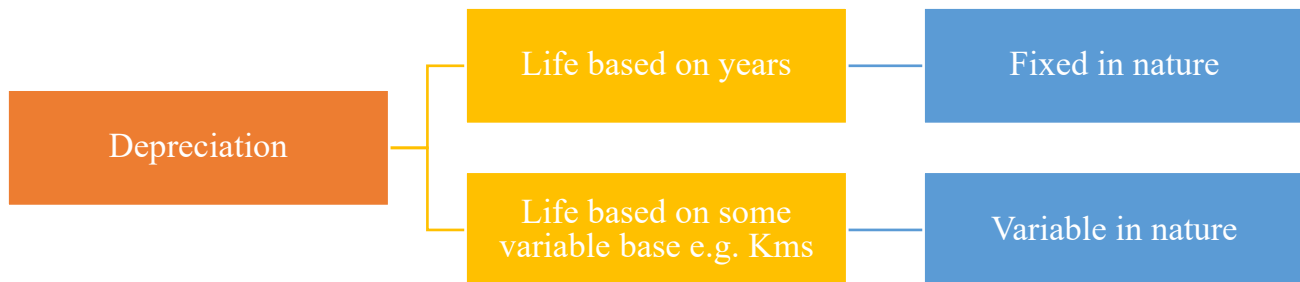


**6. Cost Per Unit** =  $\frac{\text{Total cost}}{\text{Effective cost unit}}$

**7. Points To Remember (PTRs)**

(A) Petrol, oil and similar charges are always on the basis of actual km travel

**(B) Treatment of Depreciation**



**(C) Calculation of Service Cost**

Service cost of Rs. 4,000 after every 3000 kms.

If actual kms travel; = 3,500;

No. of service =  $3500/3000 = 1.17$  or 1

Cost =  $4000 \times 1 = 4,000$

If actual kms travel; = 8,500;

No. of service =  $8500/3000 = 2.83$  or 2

Cost =  $4000 \times 2 = 8,000$

(D) In case of different charges for different categories of service, use the concept of equivalent unit of services.

**Question – 1**

Mr. S owns a bus which runs according to the following schedule:

- (a) Delhi to Panchkula and back, the same day

Distance covered: 150 kms one way

Number of days run each month: 8

Seating capacity occupied 90%

- (b) Delhi to Mathura and back, the same day

Distance covered: 120 kms one way

Number of days run each month: 10

Seating capacity occupied 85%

- (c) Delhi to Alwar and back, the same day

Distance covered: 270 kms one way

Number of days run each month: 6

Seating capacity occupied 100%

- (d) Following are the other details:

|                                    |                  |
|------------------------------------|------------------|
| Cost of the bus                    | ₹ 6,00,000       |
| Salary of the driver               | ₹ 2,800 p.m.     |
| Salary of the conductor            | ₹ 2,200 p.m.     |
| Salary of the part-time accountant | ₹ 200 p.m.       |
| Insurance of the bus               | ₹ 4,800 p.a.     |
| Diesel consumption 4 kms per litre | ₹ 6 per litre    |
| Road tax                           | ₹ 1,500 p.a.     |
| Lubricant oil                      | ₹ 10 per 100 kms |
| Permit fee                         | ₹ 315 p.m.       |
| Repairs and maintenance            | ₹ 1,000 p.m.     |
| Depreciation of the bus            | @ 20%p.a.        |
| Seating capacity of the bus        | 50 persons       |

Passenger tax is 20% of the total takings. Calculate the bus fare to be charges from each passenger to earn a profit of 30% on total takings. The fares are to be indicated per passenger for the journeys:

- (i) Delhi to Panchkula; (ii) Delhi to Mathura; and (iii) Delhi to Alwar

**Solution****Calculation of Passenger Kms**

|                   | No. | × | Kms          | × | Passenger | = | Passenger Kms |
|-------------------|-----|---|--------------|---|-----------|---|---------------|
| Delhi – Panchkula | 1   | × | 150 × 2 × 8  | × | 50 × 90%  | = | 1,08,000      |
| Delhi – Mathura   | 1   | × | 120 × 2 × 10 | × | 50 × 85%  | = | 1,02,000      |
| Delhi – Alwar     | 1   | × | 270 × 2 × 6  | × | 50 × 100% | = | 1,62,000      |
|                   |     |   |              |   | Total     |   | 3,72,000      |

Kms travel =  $(1 \times 150 \times 2 \times 8) + (1 \times 120 \times 2 \times 10) + (1 \times 270 \times 2 \times 6) = 8,040$  kms

## Statement of Operating Cost

| Particulars                                       | Amount (₹)  |
|---------------------------------------------------|-------------|
| <b>Fixed Cost:</b>                                |             |
| Depreciation $(6,00,000 \times 20\% \times 1/12)$ | 10,000      |
| Driver salary                                     | 2,800       |
| Conductor Salary                                  | 2,200       |
| Accountant salary                                 | 200         |
| Insurance $(4,800 \times 1/12)$                   | 400         |
| Road Tax $(1,500 \times 1/12)$                    | 125         |
| Permit Fee                                        | 315         |
| Total Fixed Cost (A)                              | 16,040      |
| <b>Variable Cost:</b>                             |             |
| Diesel $[8,040 \times (6/4)]$                     | 12,060      |
| Lubricating oil $[8,040 \times (10/100)]$         | 804         |
| Total Variable Cost (B)                           | 12,864      |
| <b>Maintenance Cost:</b>                          |             |
| Repair & Maintenance                              | 1,000       |
| Total Maintenance Cost (B)                        | 1,000       |
| Total Cost (A + B + C)                            | 29,904      |
| (+) Passenger tax $(59,808 \times 20\%)$          | 11,962      |
| (+) Profit $(59,808 \times 30\%)$                 | 17,942      |
| Total Takings $(29,904 \div 50\%)$                | 59,808      |
| Effective Passenger km                            | 3,72,000    |
| <b>Takings per effective passenger km</b>         | <b>0.16</b> |

## Fares per passenger for the journey

|                    |                               |
|--------------------|-------------------------------|
| Delhi to Panchkula | = $150 \times 0.16 = ₹ 24$    |
| Delhi to Mathura   | = $120 \times 0.16 = ₹ 19.20$ |
| Delhi to Alwar     | = $270 \times 0.16 = ₹ 43.20$ |

## Question – 2

SK is a public school having five buses each plying on different directions for the transport of its school students. In view of a large number of students available of the bus service the buses work two shifts daily both in the morning and in afternoon. The buses are garaged in the school. The work load of the students has been so arranged that in the morning the first trip picks up the senior students and the second trip plying an hour later picks up the junior students. Similarly, in the afternoon the first trip picks the junior students and an hour later the second trip takes the senior student home.

The distance travelled by each bus one way is 8 kms. The school works 25 days in a month and remains closed for the vacation in May, June and December. Bus fee however is payable by the students for all the 12 months of the year.

The details of expenses for a year are as under:

|                                                                                |                            |
|--------------------------------------------------------------------------------|----------------------------|
| Drivers' salary                                                                | ₹ 450 per month per driver |
| Cleaners' salary                                                               | ₹ 350 per month            |
| (Salary payable for 12 months and one cleaner employed for all the five buses) |                            |
| License fee, taxes etc.                                                        | ₹ 860 per bus per annum    |
| Insurance                                                                      | ₹ 1,000 per bus p.a.       |
| Repairs and maintenance                                                        | ₹ 3,500 per bus p.a.       |
| Purchase price of bus (life 12 years)                                          | ₹ 1,50,000 each            |
| Scrap value                                                                    | ₹ 30,000                   |
| Diesel cost                                                                    | ₹ 2.00 per litre           |

Each bus gives an average mileage of 4 km per liter of diesel.

Seating capacity of each bus is 50 students. The seating capacity is fully occupied during the whole year. Students picked up and dropped within a range upto 4 km of distance from the school are charged half fare and fifty percent of students travelling in each trip are in this category. Ignore interest since the charges are to be based on average cost.

You are required to:

- Prepare a statement showing the expenses of operating a single bus and the fleet of five buses for a year.
- Work out the average cost per student per month in respect of
  - Students coming from a distance of upto 4 km from the school and
  - Students coming from a distance of beyond 4 km from the school

### Solution

#### (a) Statement of Cost

| Particulars          | 1 bus                                  | 5 buses                    |
|----------------------|----------------------------------------|----------------------------|
| Driver salary        | $450 \times 12 = 5,400$                | $5,400 \times 5 = 27,000$  |
| Cleaner salary       | $4,200 \div 5 = 840$                   | $350 \times 12 = 4,200$    |
| License etc.         | 860                                    | $860 \times 5 = 4,300$     |
| Insurance            | 1,000                                  | $1,000 \times 5 = 5,000$   |
| Repair & Maintenance | 3,500                                  | $3,500 \times 5 = 17,500$  |
| Depreciation         | $(1,50,000 - 30,000) \div 12 = 10,000$ | $10,000 \times 5 = 50,000$ |
| Diesel (w.n. – 1)    | 7,200                                  | $7,200 \times 5 = 36,000$  |
| <b>Total</b>         | <b>28,800</b>                          | <b>1,44,000</b>            |

#### Working note 1

Number of round trips in a day = 8

Total kms in a day =  $8 \times 8 = 64$

Total kms per annum =  $64 \times 25 \times 9 = 14,400$

Diesel cost per annum per bus =  $14,400 \times (2/4) = ₹ 7,200$

(b) Number of students from whom bus fee is receivable = 100

Half fee students per bus =  $100 \times 50\% = 50$

Full fee students per bus =  $100 \times 50\% = 50$

Full fee students equivalent to half fee students =  $50 \times 2 = 100$

Total equivalent half fee students =  $50 + 100 = 150$

Total cost per bus per annum = ₹ 28,800

Total cost per bus per month =  $28,800 \div 12 = ₹ 2,400$

Total cost per month per half fee students =  $2,400 \div 150 = ₹ 16$

Total cost per month per full fee students =  $16 \times 2 = ₹ 32$

### Question – 3

A company is considering three alternative proposals for conveyance facilities for its sales personnel who have to do considerable travelling, approximately 20,000 kms every year. The proposal are as follows:

- (i) Purchase and maintain its own fleet of cars. The average cost of a car is ₹ 1,00,000
- (ii) Allow the executive use his own car and reimburse expenses at the rate of ₹ 1.60 paise per kilometer and also bear insurance costs.
- (iii) Hire cars from an agency at ₹ 20,000 per year per car. The company will have to bear costs of petrol, taxes and tyres.

The following further details are available:

Petrol ₹ 0.60 per km

Repairs and maintenance ₹ 0.20 per km

Tyre ₹ 0.12 per km

Insurance ₹ 1,200 per car per annum;

Taxes ₹ 800 per car per annum

Life of the car: 5 years with annual mileage of 20,000 kms

Resale value: ₹ 20,000 at the end of the fifth year

Work out the relative costs of three proposals and rank them.

### Solution

| Particulars                                 | Own car | Executive car | Hire car |
|---------------------------------------------|---------|---------------|----------|
| <b>Fixed Cost:</b>                          |         |               |          |
| Depreciation $[(1,00,000 - 20,000) \div 5]$ | 16,000  | -             | -        |
| Insurance                                   | 1,200   | 1,200         | -        |
| Taxes                                       | 800     | -             | 800      |
| Hire charges                                | -       | -             | 20,000   |
| Total Fixed cost                            | 18,000  | 1,200         | 20,800   |
| Kms travel                                  | 20,000  | 20,000        | 20,000   |
| Fixed cost per km (A)                       | 0.90    | 0.06          | 1.04     |

| Particulars                      | Own car     | Executive car | Hire car    |
|----------------------------------|-------------|---------------|-------------|
| <b>Variable Cost per km:</b>     |             |               |             |
| Petrol                           | 0.60        | -             | 0.60        |
| Repair & Maint.                  | 0.20        | -             | -           |
| Tyres                            | 0.12        | -             | 0.12        |
| Reimbursement                    | -           | 1.60          | -           |
| Total variable cost per km (B)   |             |               |             |
| <b>Total cost per km (A + B)</b> | <b>1.82</b> | <b>1.66</b>   | <b>1.76</b> |
| <b>Rank</b>                      | <b>III</b>  | <b>I</b>      | <b>II</b>   |

**Question – 4**

A LMV Pvt. Ltd., operates cab/car rental service in Delhi/NCR. It provides its service to the offices of Noida, Gurugram and Faridabad. At present it operates CNG fueled cars but it is also considering to upgrade these into Electric vehicles (EV). The details related with the owning of CNG & EV propelled cars are as tabulated below:

| Particulars                                   | CNG Car  | EV Car            |
|-----------------------------------------------|----------|-------------------|
| Car purchase price (₹)                        | 9,20,000 | 15,20,000         |
| Govt. subsidy on purchase of car (₹)          | -        | 1,50,000          |
| Life of the car                               | 15 years | 10 years          |
| Residual value (₹)                            | 95,000   | 1,70,000          |
| Mileage                                       | 20 km/kg | 240 km per charge |
| Electricity consumption per full charged      | -        | 30 Kwh            |
| CNG cost per Kg (₹)                           | 60       | -                 |
| Power cost per Kwh (₹)                        | -        | 7.60              |
| Annual Maintenance cost (₹)                   | 8,000    | 5,200             |
| Annual insurance cost (₹)                     | 7,600    | 14,600            |
| Tyre replacement cost in every 5 year (₹)     | 16,000   | 16,000            |
| Battery replacement cost in every 8 years (₹) | 12,000   | 5,40,000          |

Apart from the above, the following are the additional information:

| Particulars                                         |             |
|-----------------------------------------------------|-------------|
| Average distance covered by a car in a month        | 1,500 km    |
| Driver's salary (₹)                                 | 20,000 p.m. |
| Garage rent per car (₹)                             | 4,500 p.m.  |
| Share of office and administration cost per car (₹) | 1,500 p.m.  |

You have been approached by the management of A LMV Pvt. Ltd. for consultation on the two options of operating the cab service. The expected questions that may be asked by the management are as follows:

**Question - 1**

What would be the depreciable value of CNG car and EV car respectively?

- (a) ₹ 13,50,000 and ₹ 14,40,000
- (b) ₹ 15,20,000 and ₹ 8,25,000
- (c) ₹ 8,25,000 and ₹ 14,40,000
- (d) ₹ 8,25,000 and ₹ 12,00,000

**Question – 2**

What would be the monthly cost of fuel and electricity for an CNG and EV care respectively?

- (a) ₹ 4,500 and ₹ 1,425
- (b) ₹ 1,500 and ₹ 4,500
- (c) ₹ 1,525 and ₹ 1,450
- (d) ₹ 1,525 and ₹ 1,425

**Question – 3**

What would be the total cost to be incurred for replacement of tyres for CNG and EV care respectively?

- (a) ₹ 32,000 and ₹ 24,000
- (b) ₹ 12,000 and ₹ 32,000
- (c) ₹ 32,000 and ₹ 16,000
- (d) ₹ 16,000 and ₹ 12,000

**Question – 4**

What would be the total cost to be incurred for replacement of battery for CNG and EV car respectively?

- (a) ₹ 5,40,000 and ₹ 12,000
- (b) ₹ 12,000 and ₹ 5,40,000
- (c) ₹ 2,00,000 and ₹ 12,000
- (d) ₹ 1,00,000 and ₹ 2,00,000

**Question – 5**

What would be the operating cost of vehicle per month per car for both CNG and EV options?

- (a) ₹ 36,627.78 and ₹ 43,708.33
- (b) ₹ 36,627.78 and ₹ 48,523.26
- (c) ₹ 48,523.26 and ₹ 28,150.29
- (d) ₹ 48,523.26 and ₹ 28,510.29

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| D | A | C | B | A |



**Question – 5**

Coal is transported from two mines X & Y and unloaded at plots in a railway station. X is at distance of 15 kms and Y is at a distance of 20 kms from the rail head plots. A fleet of lorries having carrying capacity of 4 tonnes is used to transport coal from the mines. Records reveal that average speed of the lorries is 40 kms per hour when running and regularly take 15 minutes to unload at the rail head. At Mine X average loading time is 30 minutes per load, while at mine Y average loading time is 25 minutes per load.

Additional information:

Drivers' wages, depreciation, insurance and taxes, etc. ₹ 12 per hour

Operated Fuel, oil tyres, repairs and maintenance, etc. ₹ 1.60 per km

You are required to prepare a statement showing the cost per tonne kilometer of carrying coal from each mine 'X' and 'Y'.

**Solution****Calculation of Ton Kms**

|              | <u>No.</u> | <u>×</u> | <u>Kms</u> | <u>×</u> | <u>Ton</u> | <u>=</u> | <u>Ton Kms</u> |
|--------------|------------|----------|------------|----------|------------|----------|----------------|
| Plot to X    | 1          | ×        | 15         | ×        | 0          | =        | 0              |
| X to plot    | 1          | ×        | 15         | ×        | 4          | =        | 60             |
| <b>Total</b> |            |          |            |          |            | =        | <u>60</u>      |

$$\text{Kms travel} = (1 \times 15) + (1 \times 15) = 30$$

**Calculation of Ton Kms**

|              | <u>No.</u> | <u>×</u> | <u>Kms</u> | <u>×</u> | <u>Ton</u> | <u>=</u> | <u>Ton Kms</u> |
|--------------|------------|----------|------------|----------|------------|----------|----------------|
| Plot to Y    | 1          | ×        | 20         | ×        | 0          | =        | 0              |
| Y to Plot    | 1          | ×        | 20         | ×        | 4          | =        | 80             |
| <b>Total</b> |            |          |            |          |            | =        | <u>80</u>      |

$$\text{Kms travel} = (1 \times 20) + (1 \times 20) = 40$$

**Statement to time (in minutes)**

| <b>Particulars</b> | <b>X</b>                 | <b>Y</b>                 |
|--------------------|--------------------------|--------------------------|
| Travel time        | $30 \times (60/40) = 45$ | $40 \times (60/40) = 60$ |
| Loading time       | 30                       | 25                       |
| Unloading time     | 15                       | 15                       |
| <b>Total time</b>  | <b>90</b>                | <b>100</b>               |

**Statement of operating cost**

| <b>Particulars</b> | <b>X</b>                 | <b>Y</b>                  |
|--------------------|--------------------------|---------------------------|
| Fixed cost         | $90 \times (12/60) = 18$ | $100 \times (12/60) = 20$ |
| Variable cost      | $1.60 \times 30 = 48$    | $1.60 \times 40 = 64$     |

|                       |      |      |
|-----------------------|------|------|
| Total Cost            | 66   | 84   |
| Ton-Kms               | 60   | 80   |
| Total cost per ton-km | 1.10 | 1.05 |

**Question – 6**

SK Transport Ltd. charges ₹ 90 per ton for its 6 tons truck lorry load from city 'A' to city 'B'. The charges for the return journey are ₹ 84 per ton. No concession or reduction in these rates is made for any delivery of goods at intermediate station 'C'. In January, the truck made 12 outward journeys for city 'B' with full load out of which 2 ton were unloaded twice in the way at city 'C'. The truck carried a load of 8 tons in its return journey for 5 times but once caught by police and ₹ 1,200 was paid as fine. For the remaining trips the truck carried full load out of which all the goods on load were unloaded once at city 'C'.

The distance from city 'A' to city 'C' and city 'B' are 140 kms and 300 kms respectively. Annual fixed cost and maintenance charges are ₹ 60,000 and ₹ 12,000 respectively. Running charges spent during January, are ₹ 2,944. You are required to find out the cost per absolute ton-km and the profit for January.

**Solution****Calculation of Ton Kms**

|              | No. | × | Kms | × | Ton | = | Ton Kms       |
|--------------|-----|---|-----|---|-----|---|---------------|
| A to B       | 10  | × | 300 | × | 6   | = | 18,000        |
| A to C       | 2   | × | 140 | × | 6   | = | 1,680         |
| C to B       | 2   | × | 160 | × | 4   | = | 1,280         |
| B to A       | 5   | × | 300 | × | 8   | = | 12,000        |
| B to C       | 1   | × | 160 | × | 6   | = | 960           |
| C to A       | 1   | × | 140 | × | 0   | = | 0             |
| B to A       | 6   | × | 300 | × | 6   | = | 10,800        |
| <b>Total</b> |     |   |     |   |     | = | <b>44,720</b> |

**Statement of operating cost**

| Particulars                       | Amount (₹) |
|-----------------------------------|------------|
| Fixed charges (60,000 ÷ 12)       | 5,000      |
| Maintenance charges (12,000 ÷ 12) | 1,000      |
| Running charges                   | 2,944      |
| Total Cost                        | 8,944      |
| Ton-Km                            | 44,720     |
| Total cost per ton-km             | 0.20       |

## Statement of Profit

| Particulars            |                                 | Amount (₹) |
|------------------------|---------------------------------|------------|
| <b>Revenue:</b>        |                                 |            |
| Outward                | $10 \times 6 \times 90 = 5,400$ |            |
|                        | $2 \times 6 \times 90 = 1,080$  |            |
| Return                 | $5 \times 8 \times 84 = 3,360$  |            |
|                        | $1 \times 6 \times 84 = 504$    |            |
|                        | $6 \times 6 \times 84 = 3,024$  | 13,368     |
| Less: Total cost       |                                 | (8,944)    |
| Less: Fine & Penalties |                                 | (1,200)    |
|                        | Profit                          | 3,224      |

## Question – 7

A transport company has a fleet of three trucks of 10 tonnes, capacity each plying in different directions for transport of customer goods. The trucks run loaded with goods and return empty. The distance travelled, number trips made and the load carried per day by each truck are as under:

| Truck No. | One way distance (Km) | No. of trips per day | Load carried per trip per day (tonnes) |
|-----------|-----------------------|----------------------|----------------------------------------|
| 1         | 16                    | 4                    | 6                                      |
| 2         | 40                    | 2                    | 9                                      |
| 3         | 30                    | 3                    | 8                                      |

The analysis of maintenance cost and the total distance traveled during the last two years is as under:

| Year | Total distance traveled | Maintenance cost ₹ |
|------|-------------------------|--------------------|
| 1    | 1,60,200                | 46,050             |
| 2    | 1,56,700                | 45,175             |

The following are the details of expenses for the last year under review:

|                          |                                                                         |
|--------------------------|-------------------------------------------------------------------------|
| Diesel                   | : ₹ 10 per litre. Each liter gives 4 km per litre of diesel on average  |
| Driver Salary            | : ₹ 2,000 per month                                                     |
| License and taxes        | : ₹ 5,000 per annum per truck                                           |
| Insurance                | : ₹ 5,000 per annum for all 3 vehicles                                  |
| Purchase price per truck | : ₹ 3,00,000, Life 10 years, Scrap value at the end of life is ₹ 10,000 |
| Oil and sundries         | : ₹ 25 per 100 km run                                                   |
| General overhead         | : ₹ 11,084 per annum                                                    |

The vehicles operate 24 days per month on an average

Required:

- Prepare an Annual Cost Statement covering the fleet of three vehicles
- Calculate the cost per Km run
- Determine the freight rate per tonne km to yield a profit of 10% on freight

**Solution****Calculation of Ton Kms**

| Truck No.    | No. | × | Kms              | × | Ton | = | Ton Kms         |
|--------------|-----|---|------------------|---|-----|---|-----------------|
| 1            | 1   | × | 16 × 4 × 24 × 12 | × | 6   | = | 1,10,592        |
| 1            | 1   | × | 16 × 4 × 24 × 12 | × | 0   | = | 0               |
| 2            | 1   | × | 40 × 2 × 24 × 12 | × | 9   | = | 2,07,360        |
| 2            | 1   | × | 40 × 2 × 24 × 12 | × | 0   | = | 0               |
| 3            | 1   | × | 30 × 3 × 24 × 12 | × | 8   | = | 2,07,360        |
| 3            | 1   | × | 30 × 3 × 24 × 12 | × | 0   | = | 0               |
| <b>Total</b> |     |   |                  |   |     | = | <u>5,25,312</u> |

Total kms travelled = 1,34,784 kms

**(a) Statement of Operating Cost**

| Particulars                                                           | Amount (₹)      |
|-----------------------------------------------------------------------|-----------------|
| <b>Fixed Cost:</b>                                                    |                 |
| Driver salary (2,000 × 12 × 3)                                        | 72,000          |
| License and tax (5,000 × 3)                                           | 15,000          |
| Insurance                                                             | 5,000           |
| Depreciation $\left[ \frac{(3,00,000 - 10,000)}{10} \times 3 \right]$ | 87,000          |
| General overheads                                                     | 11,084          |
| <b>Total Fixed Cost (A)</b>                                           | <b>1,90,084</b> |
| <b>Variable Cost:</b>                                                 |                 |
| Diesel [1,34,784 × (10/4)]                                            | 3,36,960        |
| Oil & Sundries [1,34,784 × (25/100)]                                  | 33,696          |
| <b>Total Variable Cost (B)</b>                                        | <b>3,70,656</b> |
| <b>Maintenance Cost:</b>                                              |                 |
| Maintenance Cost (working note – 1)                                   | 39,696          |
| <b>Total Maintenance Cost (B)</b>                                     | <b>39,696</b>   |
| <b>Total Cost (A + B + C)</b>                                         | <b>6,00,436</b> |

**Working Note 1**

$$\text{Variable maintenance cost per km} = \frac{\text{Difference in total cost}}{\text{Difference in kms}} = \frac{46,050 - 45,175}{1,60,200 - 1,56,700} = ₹ 0.25$$

$$\text{Fixed maintenance cost} = \text{Total cost} - \text{Variable cost} = 45,175 - (1,56,700)(0.25) = ₹ 6,000$$

$$\text{Maintenance cost} = \text{Fixed cost} + \text{variable cost} = 6,000 + (1,34,784)(0.25) = ₹ 39,696$$

$$(b) \text{ Total cost per km} = \frac{\text{Total cost}}{\text{Kms}} = \frac{6,00,436}{1,34,784} = ₹ 4.45$$

$$(c) \text{ Cost per ton-km } \left( \frac{6,00,436}{1,34,784} \right) = 1.143$$

$$\text{Profit per ton-km } (1.27 \times 10\%) = 0.127$$

$$\text{Freight per ton-km } (1.143 \div 90\%) = 1.270$$

**Question – 8**

Due to frequent power cuts and inconsistent water supply from the Electricity Board, Anju Manufacturing Pvt. Ltd. — a mid-sized engineering goods manufacturer based in Pune — decided to set up its own captive power generation plant. The plant is critical for ensuring uninterrupted production across its two major units:

- Unit 1: Produces industrial machinery components (highly power- intensive operations)
- Unit 2: Manufactures agricultural equipment parts (moderate power load)

The factory operations heavily depend on continuous electric and water supply for running CNC machines, foundry operations, welding units, and assembly lines. Any electricity failure leads to major production delays, idle labour costs, and penalties from clients due to late deliveries.

After analysing the risk of depending entirely on the electricity board, the company's Board of Directors approved a proposal to invest in a dedicated steam boiler and power generation plant within the factory premises.

The following information has been taken from the records in connection with the generation of power during the month:

- Coal used 1,800 Tonnes @ ₹230 per tonne
- Freight and handling charges: 7% of the value of coal used.
- Oil: 15 Tonnes @ ₹4,500 per tonne
- Water: 80,000 litres @ ₹0.07 per litre
- Steam boiler cost ₹50,00,000 with a residual value of ₹2 lakh after a life of 10 years.
- Salaries and wages of the boiler house:
  - 15 skilled workers @ ₹1,300 per month each
  - 30 semi-skilled workers @ 1100 per month each
  - 50 unskilled workers @ ₹900 per month each
- Recovery on account of sale of ashes: 150 tonnes ₹60 per tonne.
- Salary and wages of generating station:
  - 60 skilled workers @ ₹1,200 per month each
  - 20 semi-skilled workers @ 1000 per month each
  - 30 unskilled workers @ ₹850 per month each
- Repairs and maintenance of generating equipment: ₹30,000
- Depreciation of generating equipment: ₹75,000
- Insurance premium on generating plant: ₹15,000
- Fuel handling and storage expenses: ₹8,000
- Rent of generating station premises: ₹12,000
- Share of administration charges: ₹65,000

- Total number of units generated (before losses): 2,00,000 units
- Normal losses in the process: 3,000 units generated.
- Electricity purchased from grid to meet emergency requirements: 5,000 units @ ₹6 per unit.
- 10% of the effective units generated are used internally by the generator department itself

Additional Information:

- The purchased electricity units are not subject to any loss adjustment.
- Emergency purchase units should be added to the effective units available for use.

You are required to calculate the total cost of electricity generated and the cost of generating electricity per unit.

### Solution

#### Operating Cost Sheet

| Particulars                                                                |          | Amount<br>(₹)   |
|----------------------------------------------------------------------------|----------|-----------------|
| <b>Boiler House Cost</b>                                                   |          |                 |
| Coal consumed (1,800 tonnes @ ₹ 230)                                       | 4,14,000 |                 |
| Freight, loading and unloading charges (7% of coal cost)                   | 28,980   |                 |
| Oil consumed (15 tonnes @ ₹ 4,500)                                         | 67,500   |                 |
| Water consumed (80,000 litres @ Re. 0.07)                                  | 5,600    |                 |
| Depreciation of steam boiler ( $\frac{50,00,000-2,00,000}{10 \times 12}$ ) | 40,000   |                 |
| Salaries and wages of boiler house workers:                                |          |                 |
| Skilled workers (15 @ ₹ 1,300)                                             | 19,500   |                 |
| Semi-skilled workers (30 @ 1,100)                                          | 33,000   |                 |
| Unskilled workers (50 @ ₹ 900)                                             | 45,000   |                 |
| Less: Recovery from sale of ash (150 tonnes @ ₹ 60)                        | (9,000)  |                 |
| <b>Total Cost (I)</b>                                                      |          | <b>6,44,580</b> |
| <b>Generating Station Costs</b>                                            |          |                 |
| Salaries and wages of generating station staff:                            |          |                 |
| Skilled workers (60 @ ₹ 1,200)                                             | 72,000   |                 |
| Semi-skilled workers (20 @ 1,000)                                          | 20,000   |                 |
| Unskilled workers (30 @ ₹ 850)                                             | 25,500   |                 |
| Repairs and maintenance of generating equipment                            | 30,000   |                 |
| Depreciation of generating equipment                                       | 75,000   |                 |
| Insurance premium on generating plant                                      | 15,000   |                 |
| Fuel handling and storage expenses                                         | 8,000    |                 |

|                                                     |        |                 |
|-----------------------------------------------------|--------|-----------------|
| Rent of generating station premises                 | 12,000 |                 |
| Share of administration and general expenses        | 65,000 |                 |
| <b>Total Cost (II)</b>                              |        | <b>3,22,500</b> |
| <b>Other Costs</b>                                  |        |                 |
| Electricity purchased from grid (5,000 units @ ₹ 6) | 30,000 |                 |
| <b>Total Cost (III)</b>                             |        | <b>30,000</b>   |

|                                                                     |  |                       |
|---------------------------------------------------------------------|--|-----------------------|
| <b>Total Cost of Electricity Generated and Purchased (I+II+III)</b> |  | <b>9,97,080</b>       |
| <b>Units Available (W.N.)</b>                                       |  | <b>1,81,800 units</b> |
| <b>Cost per unit</b>                                                |  | <b>5.48 per unit</b>  |

**Working Note: Units Available**

| Particulars                                   | Units           |
|-----------------------------------------------|-----------------|
| Total Units Generated                         | 2,00,000        |
| Less: Normal loss in generation               | (3,000)         |
| Net Own Units Generated                       | 1,97,000        |
| Add: Units purchased from grid                | 5,000           |
| <b>Total Effective Units Available</b>        | <b>2,02,000</b> |
| Less: 10% Units used by generating department | (20,200)        |
| <b>Units Available for Factory Operations</b> | <b>1,81,800</b> |

**Question – 9**

SK Hospital runs a Critical Care Unit (CCU) in a hired building. CCU consists of 35 beds and 5 more beds can be added, if required.

- Rent per month - ₹ 75,000
- Supervisors (2 persons) - ₹ 25,000 per month – each
- Nurse (4 persons) - ₹ 20,000 per month – each
- Ward Boys (4 persons) - ₹ 5,000 per month – each

Doctors paid ₹ 2,50,000 per month – paid on the basis of number of patients attended and the time spent by them.

**Other expenses for the year are as follows:**

- Repair (Fixed) - ₹ 81,000
- Food to patients (variable) - ₹ 8,80,000
- Other services to patients (variable) - ₹ 3,00,000
- Laundry charges (variable) - ₹ 6,00,000
- Medicines (variable) - ₹ 7,50,000
- Other fixed expenses - ₹ 10,80,000
- Administration expenses allocated - ₹ 10,00,000

It was estimated that for 150 days in a year 35 beds are occupied and for 80 days only 25 beds are occupied.

The hospital hired 750 beds at a charge of ₹ 100 per bed per day, to accommodate the flow of patients. However, this does not exceed more than 5 extra beds over and above the normal capacity of 35 beds on any day.

You are required to:

- Calculate profit per patient day, if the hospital recovers on an average ₹ 2,000 per day from each patient.
- Find out breakeven point for the hospital.

### Solution

$$(a) \text{ Effective bed days} = (150 \times 35) + (80 \times 25) + 750 = 8,000$$

#### Statement of Profit

| Particulars                  | Amount (₹)  |
|------------------------------|-------------|
| <b>Variable Cost:</b>        |             |
| Doctor cost (2,50,000 × 12)  | 30,00,000   |
| Food to patients             | 8,80,000    |
| Other services to patients   | 3,00,000    |
| Laundry charges              | 6,00,000    |
| Medicines                    | 7,50,000    |
| Bed Charges (750 × 100)      | 75,000      |
| Total Variable Cost (A)      | 56,05,000   |
| <b>Fixed Cost:</b>           |             |
| Rent (75,000 × 12)           | 9,00,000    |
| Supervisor (25,000 × 12 × 2) | 6,00,000    |
| Nurse (20,000 × 12 × 4)      | 9,60,000    |
| Ward boys (5,000 × 12 × 4)   | 2,40,000    |
| Repair                       | 81,000      |
| Other fixed expenses         | 10,80,000   |
| Administration expenses      | 10,00,000   |
| Total Fixed Cost (B)         | 48,61,000   |
| Total Cost (A + B)           | 1,04,66,000 |
| Revenue (8,000 × 2,000)      | 1,60,00,000 |
| Profit                       | 55,34,000   |

$$\text{Profit per patient day} = \frac{55,34,000}{8,000} = ₹ 691.75$$

$$(b) \text{ Contribution} = \text{Revenue} - \text{Variable cost} = ₹ 1,60,00,000 - ₹ 56,05,000 = ₹ 1,03,95,000$$

$$\text{Contribution per patient day} = \frac{1,03,95,000}{8,000} = ₹ 1,299.375$$

$$\text{Break-even point} = \frac{\text{Fixed cost}}{\text{Contribution per patient day}} = \frac{48,61,000}{1,299.375} = 3,741 \text{ patient days}$$



**Question – 10**

RST Toll Plaza Limited built an 80-kilometer-long highway between two cities and operates a toll plaza to collect tolls from passing vehicles using the highway. The company has estimated that 50,000 light weight, 12,000 medium weight and 10,000 heavy weight vehicles will be using the highway in one month in outward journey and the same number for return journey.

As per government notification, vehicles used for medical emergencies, Members of Parliament, and essential services are exempt from toll charges. It is estimated that 10% of light weight vehicles will pass the highway for such use.

It is the policy of the company that if vehicles return within 24 hours of their outward journey, the toll fare will be reduced by 25 percent automatically. It is estimated that 30% of chargeable light weight vehicles return within the specified time frame.

The toll charges for medium weight vehicles is to be fixed as 2.5 times of the light weight vehicles and that of heavy weight vehicles as 2 times of the medium weight vehicles.

The toll and maintenance cost for a month is ₹ 59,09,090. The company requires a profit of 10% over the total cost to cover interest and other costs.

Required:

- (i) Calculate the toll rate for each type of vehicle if concession facilities are not available on the return journey.
- (ii) Calculate the toll rate that will be charged from light weight vehicles if a return journey concession facility is available, assuming that the revenue earned from light weight vehicles calculated in option (i) remains the same.

**Solution****Working Notes:**

(1) Calculation of equivalent number of light vehicles

| Type of vehicles | Monthly traffic<br>(A) | Return traffic<br>(B) | Ratio (C) | Equivalent light<br>weight<br>[(A+B)×C] |
|------------------|------------------------|-----------------------|-----------|-----------------------------------------|
| Light weight     | 45,000*                | 45,000                | 1         | 90,000                                  |
| Medium weight    | 12,000                 | 12,000                | 2.5       | 60,000                                  |
| Heavy weight     | 10,000                 | 10,000                | 5         | 1,00,000                                |
|                  |                        |                       |           | 2,50,000                                |

\*50,000 light vehicles less 10% exempted vehicles.

(2) Calculation of equivalent number of light weight vehicles

| Type of vehicles | Monthly traffic (A) | Return traffic (B)                        | Ratio (C) | Equivalent light weight [(A+B)×C] |
|------------------|---------------------|-------------------------------------------|-----------|-----------------------------------|
| Light weight     | 45,000*             | 41,625<br>(45,000 – [45,000 × 30% × 25%]) | 1         | 86,625                            |
| Medium weight    | 12,000              | 12,000                                    | 2.5       | 60,000                            |
| Heavy weight     | 10,000              | 10,000                                    | 5         | 1,00,000                          |
|                  |                     |                                           |           | 2,46,625                          |

(i) Toll rate per vehicle =  $\frac{\text{Total cost to cover}}{\text{Equivalent type of vehicles}} = \frac{(59,09,090 + 10\%)}{2,50,000} = \frac{65,00,000}{2,50,000} = ₹ 26$

Toll rate for light weight vehicle = ₹ 26

Toll rate for medium weight vehicle = ₹ 26 × 2.5 = ₹ 65

Toll rate for heavy weight vehicle = ₹ 26 × 5 = ₹ 130

(ii) Revenue from light weight vehicle in (i) above = 90,000 vehicles × ₹ 26 = ₹ 23,40,000

New toll rate to maintain the same revenue from Light weight vehicle =  $\frac{23,40,000}{86,625} = ₹ 27.01$

Rate to be charged from 13,500 light weight vehicles = 27.01 × 0.75 = ₹ 20.26

#### Toll to be charged from light weight vehicles if concession applicable

Revenue share in light vehicles = 90,000 × 26 = ₹ 23,40,000

Suppose rate is x,

then outward journey 45,000 x;

return journey (45,000 – 30% of 45,000) + 13,500(x – 0.25)

45,000x + 31,500x + 13,500(0.75x) = 23,40,000

Toll rate to be charged from light weight vehicles: 86,625x = 23,40,000

X = ₹ 27.01

Rate to be charged from 76,500 light weight vehicles @ ₹ 27.01; revenue will be ₹ 20,66,494

Rate to be charged from 13,500 light weight vehicles = 27.01 × 0.75 = 20.26

Revenue will be ₹ 2,73,506

#### Question – 11

Following are the data pertaining to SK Pvt. Ltd. for the year:

| Particulars                              | Amount (₹) |
|------------------------------------------|------------|
| Salary to Software Engineers (5 persons) | 15,00,000  |
| Salary to Project Leader (2 persons)     | 9,00,000   |
| Salary to Project Manager                | 6,00,000   |
| Repairs & Maintenance                    | 3,00,000   |
| Administration Overheads                 | 12,00,000  |

The company executes a Project PK, the details of the same are as follows:

Project duration – 6 months

One Project Leader and three Software Engineers were involved for the entire duration of the project, whereas Project Manager spends 2 months' efforts, during the execution of the Project.

Travel expenses incurred for the project – ₹ 1,87,500

Two laptops were purchased at a cost of ₹ 50,000 each, for use in the project and the life of the same is estimated to be 2 years.

Prepare project cost sheet.

### Solution

#### Project Cost Sheet

| Particulars                                                            | Amount    |
|------------------------------------------------------------------------|-----------|
| Salary of software engineer ( $15,00,000 \times 3/5 \times 6/12$ )     | 4,50,000  |
| Salary of project leader ( $9,00,000 \times \frac{1}{2} \times 6/12$ ) | 2,25,000  |
| Salary of project manager ( $6,00,000 \times 2/12$ )                   | 1,00,000  |
| Total salary                                                           | 7,75,000  |
| Travel expenses                                                        | 1,87,500  |
| Depreciation ( $50,000 \times 2 \times \frac{1}{2} \times 6/12$ )      | 25,000    |
| Overheads ( $50\% \times 7,75,000$ )                                   | 3,87,500  |
| Total Project cost                                                     | 13,75,000 |

#### Working Note:

Total overheads = 3,00,000 + 12,00,000 = ₹ 15,00,000

Total Salary = 15,00,000 + 9,00,000 + 6,00,000 = ₹ 30,00,000

Overheads recovery rate =  $\frac{\text{Overheads}}{\text{Salary}} \times 100 = \frac{15,00,000}{30,00,000} \times 100 = 50\% \text{ of salary}$

#### Question – 12

A lodging home is being run in a small hill station with 50 single rooms. The home offers concessional rates during six off-season months in a year. During this period, half of the full room rent is charged. The management's profit margin is targeted at 20% of the room rent. The following are the cost estimates and other details for the year ending 31<sup>st</sup> March, (assume a month to be of 30 days):

(a) Occupancy during the season is 80%, while in the off season is 40% only:

(b) Expenses:

₹

|                                              |          |
|----------------------------------------------|----------|
| (1) Staff Salary (excluding room attendants) | 2,75,000 |
| (2) Repairs to Buildings                     | 1,30,500 |
| (3) Laundry & Linen                          | 40,000   |
| (4) Interior and Tapestry                    | 87,500   |
| (5) Sundry Expenses                          | 95,400   |

(c) Annual depreciation is to be provided for buildings at 5% and on furniture and equipments at 15% on straight line basis:

(d) Room attendants are paid ₹ 5 per room day on the basis of occupancy of the rooms in a month

- (e) Monthly lighting charges are ₹ 120 per room, except in four months of winter when it is ₹ 30 per room and this cost is on the basis of full occupancy for a month, and
- (f) Total investment in the home is ₹ 100 lakhs of which ₹ 80 lakhs relate to buildings and balance for furniture and equipments.

You are required to compute the room rent per day both during the season and off season.

### Solution

#### Computation of Effective room days

|                                                    |   |               |
|----------------------------------------------------|---|---------------|
| Season = (50 rooms × 80 / 100) × (6 × 30) days     | = | 7,200         |
| Off-season = (50 rooms × 40 / 100) × (6 × 30) days | = | <u>3,600</u>  |
|                                                    |   | <u>10,800</u> |

#### Computation of Total Cost

|                                                                      |                 | ₹                |
|----------------------------------------------------------------------|-----------------|------------------|
| (1) Salary                                                           |                 | 2,75,000         |
| (2) Repairs                                                          |                 | 1,30,500         |
| (3) Laundry and Linen                                                |                 | 40,000           |
| (4) Interior Decoration                                              |                 | 87,500           |
| (5) Depreciation                                                     |                 |                  |
| -Building (5% of 80,00,000)                                          | 4,00,000        |                  |
| -Furniture & equipment (15% of 20,00,000)                            | <u>3,00,000</u> | 7,00,000         |
| Sundry Expenses                                                      |                 | 95,400           |
| (6) Attendant's Salary (10,800 × 5)                                  |                 | 54,000           |
| (7) Lighting Charges                                                 |                 |                  |
| - Season (7,200 days × ₹ 4) [₹ 120 p.m. means ₹ 4 per day]           |                 | 28,800           |
| - Off-Season                                                         |                 |                  |
| Winter [3,600 × 4/6] × ₹ 1 [₹ 30 p.m. means Re.1 per day] (4 months) |                 | 2,400            |
| Balance (3,600 × 2/6) × ₹ 4 (2 months)                               |                 | 4,800            |
| Total Cost                                                           |                 | <u>14,18,400</u> |

#### Computation of Total Revenue

|                                                   | ₹                |
|---------------------------------------------------|------------------|
| Total Cost                                        | 14,18,400        |
| (+) Profit (20% of revenue) (14,18,400 × 20/80) - | <u>3,54,600</u>  |
| Total Revenue                                     | <u>17,73,000</u> |

Assume Rent per room per day during Season is ₹ X & during off season is ₹ X/2

Hence, total annual revenue = 7,200 X + 3,600 [X/2] = 9,000 X

Now, 9,000 X = ₹ 17,73,000

X = 197 Hence, Rent per room per day

During Season = X = ₹ 197

During off-season = [X/2] = ₹ [197/2] = ₹ 98.50

**Question – 13**

ABC Bank is having a branch which is engaged in processing of 'Vehicle Loan' and 'Education Loan' applications in addition to other services to customers. 30% of the overhead costs for the branch are estimated to be applicable to the processing of 'Vehicle Loan' applications and 'Education Loan' applications each.

Branch is having four employees at a monthly salary of ₹ 50,000 each, exclusively for processing of Vehicle Loan applications and two employees at a monthly salary of ₹ 70,000 each, exclusively for processing of Education Loan applications.

In addition to above, following expenses are incurred by the Branch:

- Branch Manager who supervises all the activities of branch, is paid at ₹ 90,000 per month.
- Legal charges, Printing & stationery and advertising expenses are incurred at ₹ 30,000, ₹ 12,000 and ₹ 18,000 respectively for a month.
- Other expenses are ₹ 10,000 per month.

You are required to:

- Compute the cost of processing a Vehicle Loan application on the assumption that 496 Vehicle Loan applications are processed each month.
- Find out the number of Education Loan applications if the total processing cost per Education Loan Application is same as in the Vehicle loan Application as computed in (i) above.

**Solution**

| Particulars                                          | Vehicle Loan Applications (₹) | Education Loan Applications (₹) | Total (₹)       |
|------------------------------------------------------|-------------------------------|---------------------------------|-----------------|
| Employee Cost                                        | $50,000 \times 4 = 2,00,000$  | $70,000 \times 2 = 1,40,000$    | 3,40,000        |
| Apportionment of branch manager's salary             | 27,000                        | 27,000                          | 54,000          |
| Legal charges, printing & stationery and advertising | 18,000                        | 18,000                          | 36,000          |
| Other expenses                                       | 3,000                         | 3,000                           | 6,000           |
| <b>Total cost</b>                                    | <b>2,48,000</b>               | <b>1,88,000</b>                 | <b>4,36,000</b> |

$$(a) \text{ Cost of processing vehicle loan application} = \frac{\text{Total cost}}{\text{No. of applications}} = \frac{2,48,000}{496} = ₹ 500$$

$$(b) \text{ Cost of processing education loan application} = \frac{\text{Total cost}}{\text{No. of applications}}$$

$$500 = \frac{1,88,000}{\text{No. of applications}}$$

$$\text{No. of applications} = \frac{1,88,000}{500} = 376$$

**Question – 14**

MRSL Healthcare Ltd. has incurred the following expenditure during the last year for its newly launched 'COVI-19' Insurance policy:

|                                 | ₹           |
|---------------------------------|-------------|
| Office administration cost      | 48,00,000   |
| Claim management cost           | 3,80,000    |
| Employees' cost                 | 16,20,000   |
| Postage and logistics           | 32,40,000   |
| Policy issuance cost            | 29,50,000   |
| Facilities cost                 | 46,75,000   |
| Cost of marketing of the policy | 1,38,90,000 |
| Policy development cost         | 35,00,000   |
| Policy servicing cost           | 96,45,000   |
| Sales support expenses          | 32,00,000   |
| IT cost                         | ?           |

Number of Policy sold: 2,800

Total insured value of policies - ₹3,500 crores

Cost per rupee of insured value - ₹0.002

You are required to:

- Calculate the total cost for "COVID-19" Insurance policy segregating the costs into four main activities namely (a) Marketing and Sales support (b) operations (c) IT Cost and (d) Support functions.
- Calculate cost per policy

**Solution**

- Total Cost = Total insured value × Cost per rupee of insured value

$$\text{Total Cost} = ₹3,500 \text{ crore} \times 0.002$$

$$\text{Total Cost} = ₹7,00,00,000$$

$$\text{Other Cost} + \text{IT Cost} = 7,00,00,000$$

$$4,79,00,000 + \text{IT Cost} = 7,00,00,000$$

$$\text{IT Cost} = ₹2,21,00,000$$

**Statement of Cost**

| Particulars                         | Amount             |
|-------------------------------------|--------------------|
| <b>Marketing and Sales Support:</b> |                    |
| Cost of marketing the policy        | 1,38,90,000        |
| Policy development cost             | 35,00,000          |
| Sales support expenses              | 32,00,000          |
| <b>Total (A)</b>                    | <b>2,05,90,000</b> |
| <b>Operations Cost:</b>             |                    |
| Claim management cost               | 3,80,000           |
| Policy issuance cost                | 29,50,000          |
| Policy servicing cost               | 96,45,000          |
| <b>Total (B)</b>                    | <b>1,29,75,000</b> |

|                                   |                    |
|-----------------------------------|--------------------|
| <b>IT Cost:</b>                   |                    |
| IT Cost                           | 2,21,00,000        |
| <b>Total (C)</b>                  | <b>2,21,00,000</b> |
| <b>Support Function:</b>          |                    |
| Office administration cost        | 48,00,000          |
| Employees' cost                   | 16,20,000          |
| Postage and logistics             | 32,40,000          |
| Facilities cost                   | 46,75,000          |
| <b>Total (D)</b>                  | <b>1,43,35,000</b> |
| <b>Total Cost (A + B + C + D)</b> | <b>7,00,00,000</b> |
| Number of Policies                | 2,800              |
| <b>Cost per policy</b>            | <b>25,000</b>      |